

Vision AI System

User Manual and Configuration Guide

Walk-in Greeting AI MVP - 2026-07-04

This document explains how to deploy, configure, and operate the Vision AI System MVP. The first module, Walk-in Greeting AI, connects to CCTV/IP cameras, detects faces, recognizes registered staff or customers, and creates real-time greeting events. Screenshot examples use demo data from the MVP interface.

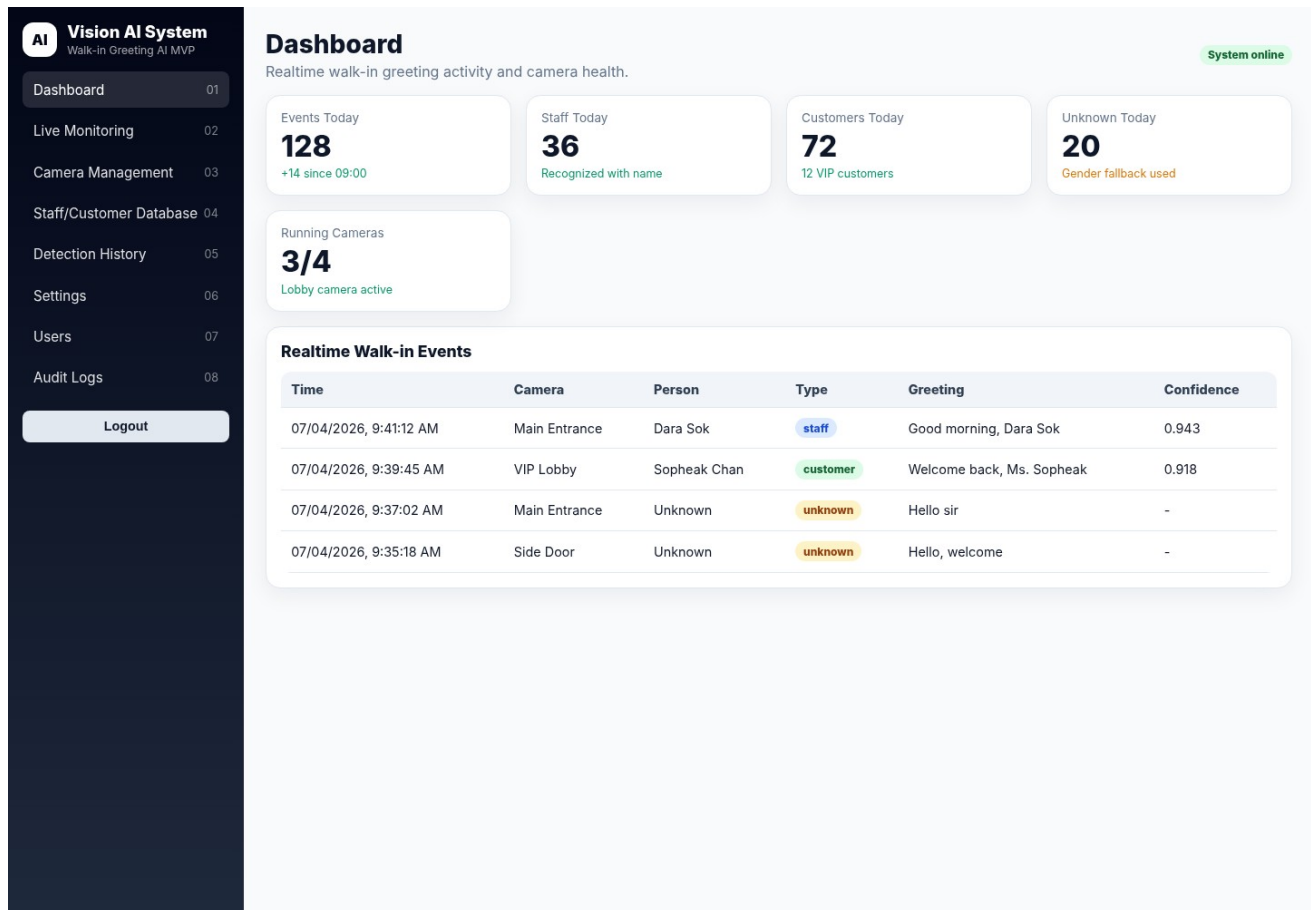


Figure 1 - Dashboard showing real-time walk-in statistics and camera health.

Contents

- Deployment quick start
- First login and user roles
- Camera configuration
- Person registration and face enrollment
- Live monitoring and detection history
- AI settings, greeting templates, privacy, backup, and troubleshooting

1. System overview

The application is a modular Vision AI platform. The current MVP focuses on the Walk-in Greeting AI module. Future modules can be added for PPE detection, queue counting, intrusion detection, attendance, people counting, vehicle plate recognition, or branch analytics.

Service	Purpose
frontend/	Nuxt web UI for dashboard, cameras, persons, settings, users, and logs.
backend/	FastAPI API for authentication, role access, database, events, and dashboard SSE.
ai-service/	FastAPI AI service for RTSP workers, face detection, embeddings, recognition callbacks, and greeting logic.
database/	PostgreSQL schema with pgvector for vector similarity search.
storage/	Shared local storage for person face images and event snapshots.
docker/	Production-style Docker deployment with nginx, health checks, volumes, scripts, and env template.

2. Deployment quick start

2.1 Local development stack

Use the root Docker Compose file for a simple local development run.

```
cp .env.example .env
docker compose up --build
```

URL	Purpose
http://localhost:3000	Frontend app
http://localhost:8000/docs	Backend API documentation
http://localhost:8001/docs	AI service API documentation

2.2 Production-style Docker stack

Use the deployment folder for nginx reverse proxy, health checks, named volumes, and production environment values.

```
cp docker/.env.production.example docker/.env.production
# edit docker/.env.production and replace all passwords/secrets
docker compose --env-file docker/.env.production -f docker/docker-compose.yml up -d --build
```

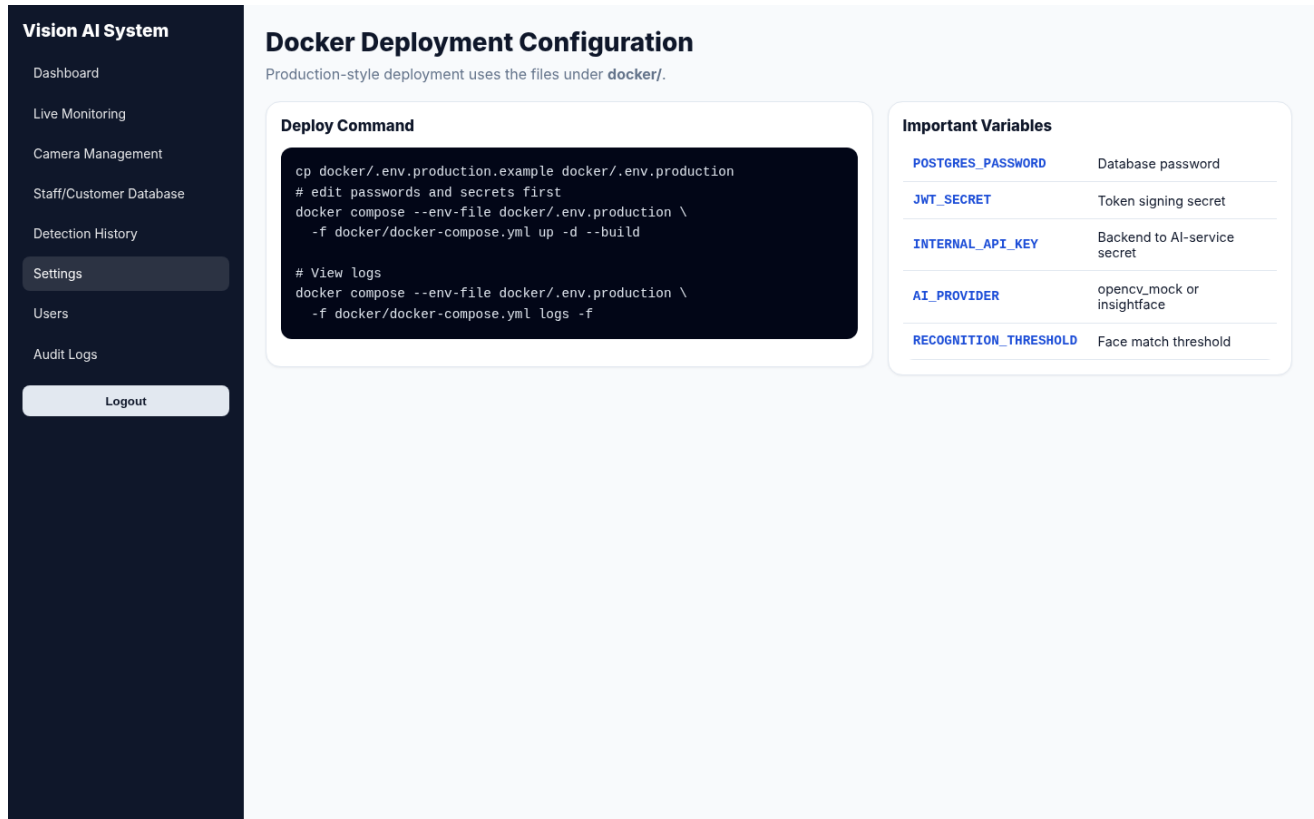


Figure 2 - Production-style Docker deployment configuration.

Endpoint	Purpose
http://localhost	Main app through nginx
http://localhost/api	Backend API through nginx
http://localhost/health	nginx health endpoint

3. First login and roles

Default account	Value
Email	admin@example.com
Password	admin123

Change the default password immediately before any real deployment. Create separate users for managers, operators, and viewers instead of sharing the Admin account.

Role	Recommended access
Admin	Full access to users, roles, cameras, persons, settings, events, and audit logs.
Manager	Dashboard, reports, detection history, and staff/customer profile management.
Operator	Live monitoring, camera operation, and limited person registration if allowed by policy.
Viewer	Read-only access to dashboard and reports.

4. Camera configuration

Open Camera Management, enter the camera name, branch, location, and RTSP URL, then add the camera. Start the worker only after the stream has been tested.

```
rtsp://username:password@camera-ip:554/stream1  
rtsp://username:password@camera-ip:554/h264/ch1/main/av_stream
```

- Use wired network where possible for entrance cameras.
- Place cameras near face height and avoid strong backlight.
- Start with 3-5 AI FPS and tune after measuring CPU/GPU usage.
- Keep RTSP credentials private and rotate passwords periodically.

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Dashboard 01
Live Monitoring 02
Camera Management 03
Staff/Customer Database 04
Detection History 05
Settings 06
Users 07
Audit Logs 08
Logout

Camera Management

Register RTSP/IP cameras and start or stop the AI worker per camera. RTSP ready

Add Camera

Name: Main Entrance Branch: Head Office Location: Lobby

RTSP URL: rtsp://user:password@192.168.1.50:554/stream1

Add Camera

Name	Branch	Location	Status	RTSP	Actions
Main Entrance	Head Office	Lobby	running	rtsp://user:***@192.168.1.50:554/stream1	Start Stop Delete
VIP Lobby	Head Office	Customer Hall	running	rtsp://user:***@192.168.1.51:554/ch01	Start Stop Delete
Side Door	Branch A	Entrance	idle	rtsp://user:***@192.168.2.20:554/live	Start Stop Delete

Figure 3 - Camera Management page for RTSP camera setup and worker start/stop actions.

5. Staff/customer registration and face enrollment

Register each person only after consent is confirmed. Add staff ID, customer ID, branch, gender, department/position, and profile details as applicable. Upload multiple clear images so the AI service can create reliable embeddings.

1. Open Staff/Customer Database.
2. Click Add Person and select Staff or Customer.
3. Enter full name, gender, branch, and optional staff/customer details.
4. Confirm consent for biometric registration.
5. Open the profile and upload 3-5 clear face images.

6. Confirm that embeddings are generated successfully.

The screenshot displays the 'Staff / Customer Face Database' interface within the 'Vision AI System'. The left sidebar contains a navigation menu with items: Dashboard (01), Live Monitoring (02), Camera Management (03), Staff/Customer Database (04), Detection History (05), Settings (06), Users (07), and Audit Logs (08). A 'Logout' button is at the bottom of the sidebar. The main content area is titled 'Staff / Customer Face Database' and includes a sub-header 'Register consented staff and customer profiles with multiple face photos.' and an 'Add Person' button. Below this is a tabbed interface with 'All', 'Staff', and 'Customers' tabs. The 'Staff' tab is active, showing a table with the following data:

Name	Type	Gender	Branch	Status	Embedding	Actions
DS Dara Sok Staff ID: STF-001	staff	male	Head Office	active	3 images	Open
SC Sopheak Chan VIP Customer	customer	female	Head Office	active	4 images	Open
KV Kosal Vann Staff ID: STF-014	staff	male	Branch A	active	2 images	Open

Figure 4 - Staff/customer database and profile image upload workflow.

6. Live monitoring

Use the Live Monitoring page to confirm camera frames, face boxes, recognition labels, and greeting outputs. Known staff/customers are greeted by name. Unknown people use sir/madam/neutral greeting based on gender confidence.

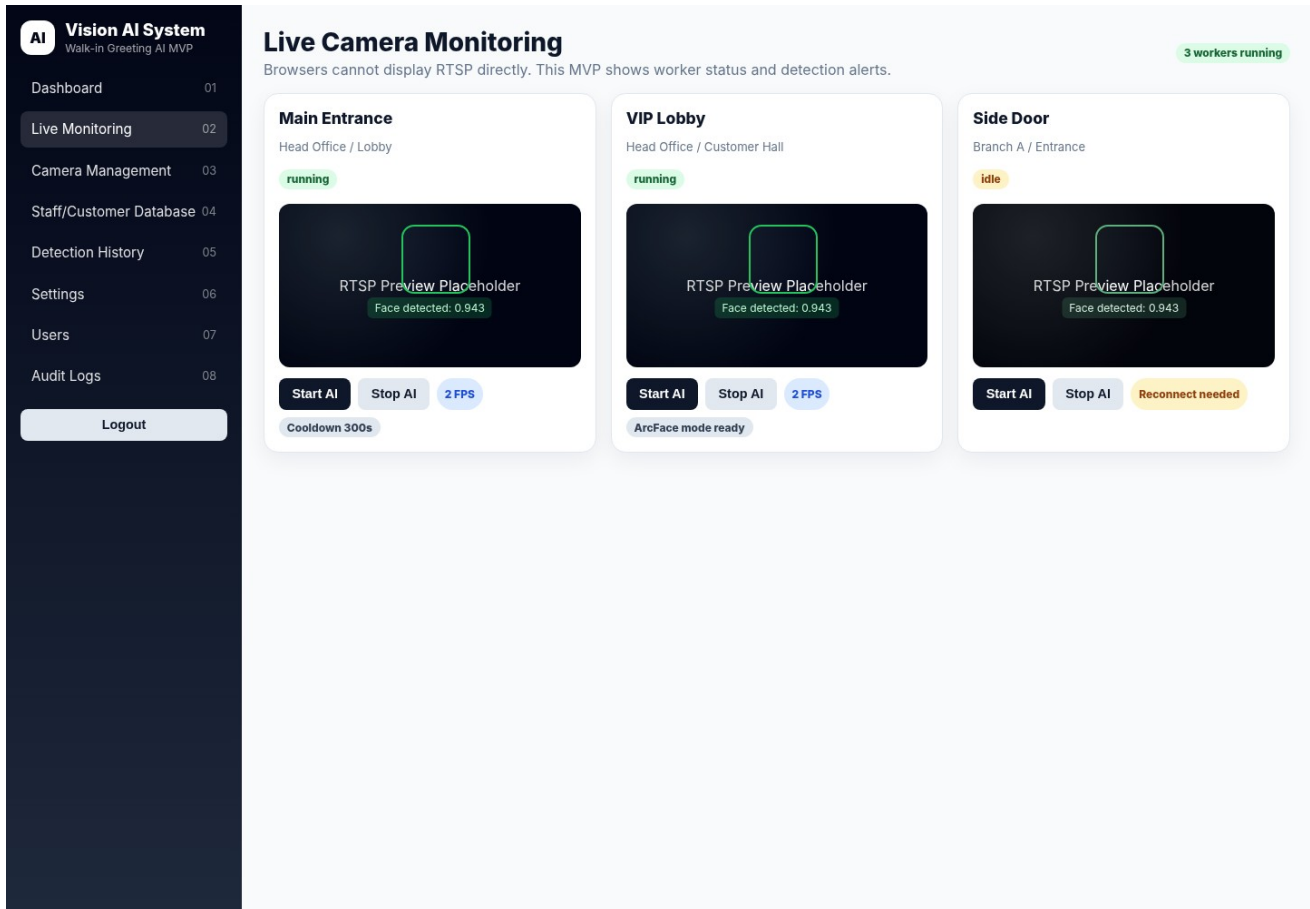


Figure 5 - Live monitoring with face box, recognition confidence, and greeting output.

- Known person greeting example: Good morning, Dara Sok.
- Unknown male estimate: Hello sir.
- Unknown female estimate: Hello madam.
- Low gender confidence: Hello, welcome.
- The cooldown setting prevents repeated greeting for the same person within the configured window.

7. Detection history and reports

Every detection event stores the timestamp, camera, snapshot, person decision, person type, confidence score, gender estimate for unknown visitors, and greeting message. Use this page for operational review and report export.

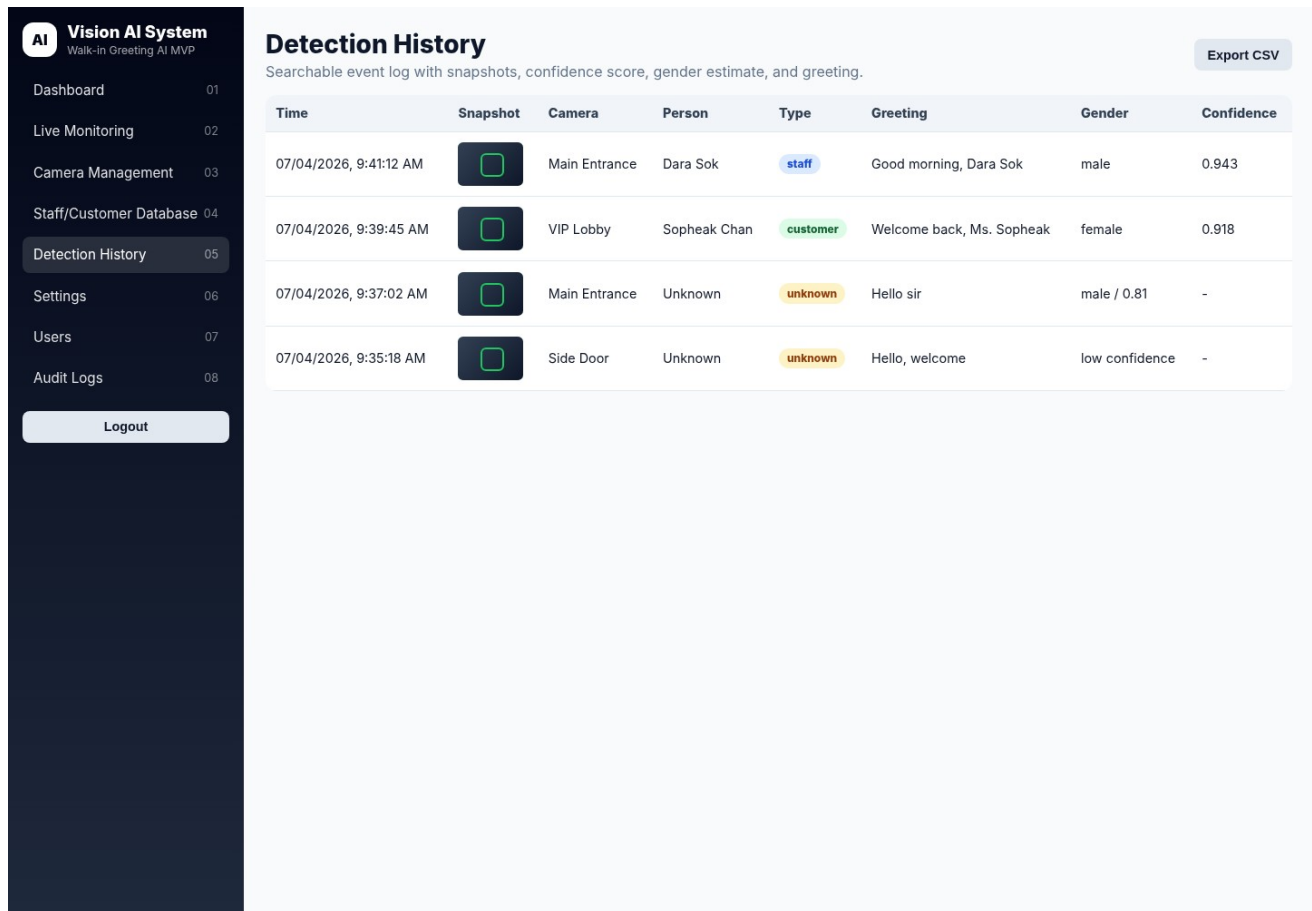


Figure 6 - Detection History page with snapshots, greeting, type, and confidence.

8. AI settings and greeting templates

Settings control recognition behavior and greeting output. Tune carefully using real branch camera footage and enrollment photos before production use.

Setting	Purpose	Typical value
recognition_threshold	Minimum vector similarity for a known face match.	0.60-0.70
greeting_cooldown_seconds	Prevents duplicate greeting for the same person.	300
gender_min_confidence	Uses neutral greeting when gender estimate is below this value.	0.75
process_fps	Frames per second processed by the AI worker.	3-8
AI_PROVIDER	opencv_mock for development or insightface for production.	opencv_mock / insightface

AI Vision AI System

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Dashboard01

Live Monitoring02

Camera Management03

Staff/Customer Database04

Detection History05

Settings06

Users07

Audit Logs08

Logout

Settings

Tune AI thresholds, greeting cooldown, and templates.

AI Settings

recognition_threshold

0.45

Minimum pgvector similarity confidence for known person recognition.

Save

greeting_cooldown_seconds

300

Avoid repeated greeting for the same person.

Save

gender_min_confidence

0.60

Use neutral greeting if unknown gender confidence is too low.

Save

Greeting Templates

Known

Good {time_of_day}, {name}

Male Unknown

Hello sir

Female Unknown

Hello madam

Neutral Unknown

Hello, welcome

Save Template

Privacy reminder: register only consented people, restrict access by role, and define retention for snapshots and event logs.

Figure 7 - AI settings and greeting template configuration.

9. Environment variables

Variable	Description
DATABASE_URL	PostgreSQL connection string used by the backend.
POSTGRES_USER / POSTGRES_PASSWORD / POSTGRES_DB	Database account and database name for Compose deployment.
JWT_SECRET	Secret used to sign application login tokens. Replace in production.
INTERNAL_API_KEY	Shared secret for backend and AI service internal calls. Replace in production.
AI_SERVICE_URL	Backend URL for calling the AI inference service.
BACKEND_URL	AI service URL for calling backend internal recognition/event APIs.
AI_PROVIDER	opencv_mock for lightweight development or insightface for production recognition.
RECOGNITION_THRESHOLD	Default threshold for face vector similarity match.
GREETING_COOLDOWN_SECONDS	Default greeting repeat cooldown.
GENDER_MIN_CONFIDENCE	Minimum gender confidence for sir/madam greeting.
STORAGE_DIR	Shared storage path for face images and snapshots.

10. Privacy and security checklist

- Collect and record consent before registering face data.

- Post signage that AI CCTV detection is active.
- Use HTTPS in production.
- Replace all default secrets and passwords.
- Restrict access by role and review user permissions regularly.
- Protect storage paths and avoid public image URLs.
- Define retention policy for event snapshots and logs.
- Delete face images and embeddings when required by policy or user request.
- Review local law and compliance requirements before production use.

11. Backup and restore

Use the included deployment scripts for PostgreSQL backup and restore.

```
# Backup
bash docker/scripts/backup-postgres.sh

# Restore
bash docker/scripts/restore-postgres.sh backups/your-backup.sql
```

12. Troubleshooting

Problem	What to check
Camera does not start	RTSP URL, camera credentials, firewall, network route, and AI-service logs.
No detections	Camera angle, lighting, process FPS, face size in frame, and worker status.
Wrong recognition	Add better enrollment photos, remove bad images, tune threshold, and verify model provider.
Dashboard does not update	Backend SSE endpoint, nginx proxy config, browser console, and backend logs.
Image upload fails	Storage volume permission, backend file route, AI service health, and internal API key.
Docker stack fails	Run docker compose ps and check logs for postgres, backend, ai-service, frontend, and nginx.

13. Important project paths

docker/docker-compose.yml	Production-style Docker Compose
docker/.env.production.example	Production env template
docker/nginx/nginx.conf	Reverse proxy configuration
database/init.sql	PostgreSQL and pgvector schema
backend/app/	FastAPI backend
ai-service/app/	AI inference service
frontend/app/	Nuxt frontend
storage/	Shared local file storage